

Gabriel Bendix

Education

The Herbert Wertheim College of Engineering, University of Florida

Gainesville, Florida

Bachelor of Science in Computer Engineering

May 2027

Relevant Coursework: Microprocessors Applications, Operating Systems, Digital Logic and Design, Circuits

Relevant Experience

Texas Instruments

Dallas, Texas

Product Marketing Engineer Intern

Sept 2026 – Aug 2026

- Developed and executed comprehensive marketing guides for both publicly released and NDA-protected internal motor driver parts, directly addressing a 16% customer loss rate in key target industries.
- Co-authored a customer application note and conducted lab experiments on the thermal performance of the DRV7308, the industry's first 650V Intelligent Power Module motor driver built on Gallium Nitride.
- Architected and implemented UI/UX changes for the Brushless DC Motor Driver product pages on TI.com.

University of Florida Computer Engineering Department

Gainesville, Florida

FPGA Hardware Pipeline Research Co-Author

Jan 2025 – Present

- Co-authored "Real-Time Edge Semantics for Drone Swarms via FPGA Perception and On-Device LLMs" (IEEE Xplore): an FPGA front end performs real-time multi-object tracking for UAV swarms, emitting a compact per-object state that a quantized on-device LLM ingests to infer swarm-level intent.
- Developed a pipeline that eliminates cloud dependence, slash uplink needs and keep latency deterministic.
- Achieved 66.7 ms end-to-end latency at ~2.9 W on a Xilinx KV260 platform, using under 2% of LUT/FF available.

The Lens Link

Gainesville, Florida

Software Development Team Lead

Sept 2025 – Jan 2026

- Led React Native mobile app MVP development with camera recognition and on-device eyewear try-on.
- Architected The Lens Link e-commerce platform using Next.js APIs, CI/CD, analytics, and adaptive marketing.

CANAC Data Processing

São Paulo, Brazil

Software Development Intern

May 2025 – July 2025

- Utilized the production data from Brazilian cane mills to develop online panels, providing harvesters with real time analytics and alerts when making logistical decisions, increasing total cane production and data accuracy by 20%.
- Leveraged SQL with Google BigQuery to analyze 500,000 data points, building interactive React dashboards with filterable climate heat maps, vector tile layers, and station pins.

Leadership Development

UF Engineering Innovation Institute

Gainesville, Florida

Program Leader

May 2024 – October 2024

- Led IGNITE, the only EII backed program promoting entrepreneurship at UF with panels, VCs, and workshops.
- Executed a campus outreach overhaul that increased awareness by 80%, building cross organization partnerships (e.g. SHPE, UF Blackstone Launchpad), establishing flagship events, and launching targeted media campaigns.

FEI University Robotics

São Bernardo, Brazil

Electrical Team Lead

May 2025 – July 2025

- Deployed a GPU accelerated computer vision pipeline with ZED 2i depth sensing camera for autonomous robots.
- Used CUDA and ROS on Ubuntu Linux to run autonomous navigation algorithms for a robotics competition.

Skills & Interests

Proficient Languages: Portuguese, English, Spanish, Assembly, VHDL, Verilog, Python, C++, JavaScript, CSS

Programs: Miami Youth Symphony (Violinist), SHPE (Program Mentor), BRASA (Program Mentor)